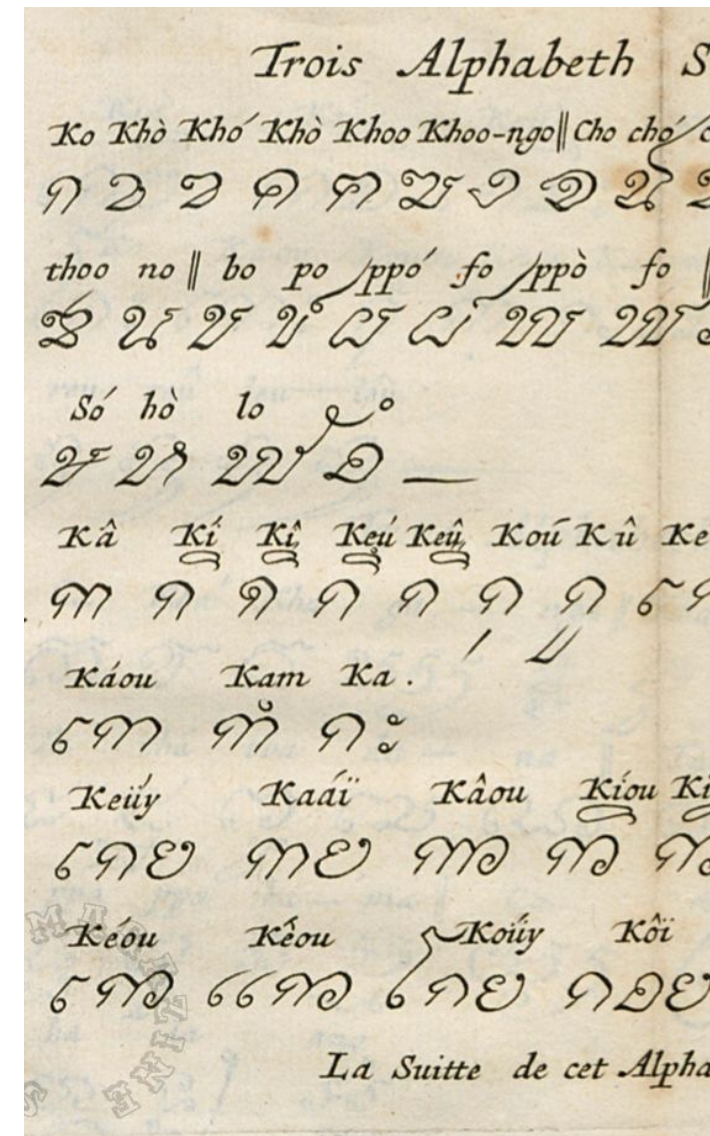


A Brief Specification of LST20 Corpus

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Outline

- Executive summary
- Agreement of usage & download
- Annotation schemes
- Statistics

Executive Summary

Executive Summary

- LST20 Corpus
 - Dataset for training fundamental Thai language processing tasks
 - Featured linguistic information
 - **Word boundaries** for word segmentation
 - **Named entities** for named entity recognition
 - **Clause boundaries** for clause segmentation
 - **Sentence boundaries** for sentence segmentation
 - **News genres** for document classification
 - CoNLL-2003 format: tab-separated columns

Words	3,162,864
Named entities	288,020
Clauses	248,962
Sentences	74,180
Distinct words	46,692
Genres	15
News articles	3,745

Available at
<https://aiforthai.in.th>

Agreement of Usage & Download

Agreement of Usage

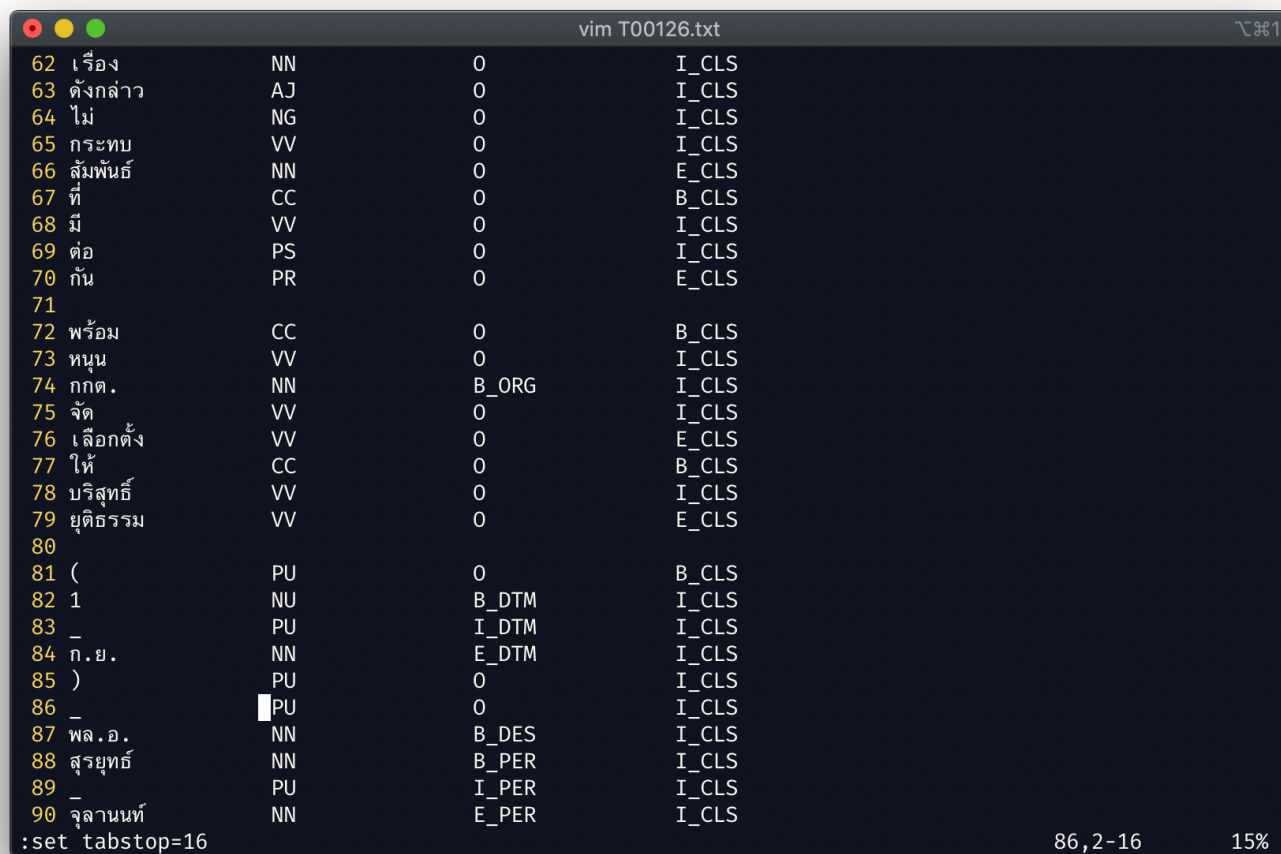
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- **Download**
 - <https://aiforthai.in.th>
 - Registration is required before downloading
 - Much more web APIs and demos are also available therein

Annotation Schemes

Format: CoNLL-2003 Style



62	เรื่อง	NN	0	I_CLS
63	ดังกล่าว	AJ	0	I_CLS
64	ไม่	NG	0	I_CLS
65	กระทบ	VV	0	I_CLS
66	สัมพันธ์	NN	0	E_CLS
67	ที่	CC	0	B_CLS
68	มี	VV	0	I_CLS
69	ต่อ	PS	0	I_CLS
70	กัน	PR	0	E_CLS
71				
72	พร้อม	CC	0	B_CLS
73	หมุน	VV	0	I_CLS
74	กกด.	NN	B_ORG	I_CLS
75	จัด	VV	0	I_CLS
76	เลือกตั้ง	VV	0	E_CLS
77	ให้	CC	0	B_CLS
78	บริสุทธิ์	VV	0	I_CLS
79	ยุติธรรม	VV	0	E_CLS
80				
81	(	PU	0	B_CLS
82	1	NU	B_DTM	I_CLS
83	-	PU	I_DTM	I_CLS
84	ก.ย.	NN	E_DTM	I_CLS
85	)	PU	0	I_CLS
86	-	PU	0	I_CLS
87	พล.อ.	NN	B_DES	I_CLS
88	สุรยุทธ์	NN	B_PER	I_CLS
89	-	PU	I_PER	I_CLS
90	จุลานนท์	NN	E_PER	I_CLS

- Four columns
 - Word
 - POS tag
 - Named entity
 - Clause boundary
- Notes
 - Each column is separated by a tab
 - Empty line marks sentence boundary

POS Tagset

Tags	Names	Brief Descriptions	Tags	Names	Brief Descriptions
AJ	Adjective	Attribute, modifier, or description of a noun	NN	Noun	Person, place, thing, abstract concept, and proper name
AV	Adverb	Word that modifies or qualifies an adjective, verb, or another adverb	NU	Number	Quantity for counting and calculation
AX	Auxiliary	Tense, aspect, mood, and voice	PA	Particle	Politeness, intention, belief, question
CC	Connector	Conjunction and relative pronoun	PR	Pronoun	Word used to refer to an element in the discourse
CL	Classifier	Class or measurement unit to which a noun or an action belongs	PS	Preposition	Location, comparison, instrument, exemplification
FX	Prefix	Inflectional (nominalizer, adjectivizer, adverbializer, and courteous verbalizer), and derivational	PU	Punctuation	Punctuation mark
IJ	Interjection	Exclamation word	VV	Verb	Action, state, occurrence, and word that forms the predicate part
NG	Negator	Word of negation	XX	Others	Unknown category

* **Green texts = content word** | **Black texts = function word** | **Red texts = undesirable (yet they still exist)**

Named Entities

Tags	Names	Descriptions
TTL	Title	Family relationship, social relationship, and permanent title
DES	Designation	Position and professional title
PER	Person	Name of a person or family
ORG	Organization	Name of organization, office, or company
LOC	Location	Name of land according to geo-political borders
BRN	Brand	Name of brand, product, and trademark
DTM	Date and time	Time or a specific period of time
MEA	Measurement	Measurement unit and quantity of things
NUM	Number	The number of a measurement unit
TRM	Terminology	Domain-specific word

- BIEO tagging convention
 - *B_tag* = beginning
 - *I_tag* = intermediate/inside
 - *E_tag* = ending
 - *O_tag* = outside
- Notes
 - BIEO convention can be converted to BIO by replacing all *E_tag*'s with *I_tag*

Green texts = personal entity | Red texts = collective entity | Black texts = referential entity

Clause Boundary

- **Definition:** clause = a part of sentence that contains at least one verb
- Syntactic clues for clause boundary
 - Subordinate connector e.g. ซึ่ง, ถ้า, ว่า
 - Cohesive marker e.g. อย่างไรก็ตาม, นอกจากนี้
 - List marker e.g. เช่น, ได้แก่, ตามลำดับ
 - Particle e.g. ครับ, นะ
 - Question adverbs e.g. อย่างไร, ไหม

Sentence Boundary

- **Definition:** sentence = (1) a group of at least one clause, or (2) a phrase acting as a topic
- Clues for sentence segmentation
 - Topic shift denoted with a cohesive marker: **break**
 - Subject shift between two adjacent clauses: **break**
 - Direct speech “.....”: **concatenate** inside quotes
 - Indirect speech: **break** when the subject changes
 - Item list: **concatenate** all items
 - Particle e.g. ครับ, นะ: **break**

Statistics

News Genres

Politics	Royal
Crime and Accident	Disaster
Economics	Urban development
Entertainment	Environment
Sports	Culture
International	Weather forecast
Science, Technology, and Education	General
Health	

- Specification
 - **Size:** 3,745 articles
 - **Sources:** Thairath, Dailynews, Manager, Matichon, Nation, Prachachat Business
 - **Duration:** JAN 2003 to DEC 2009

Corpus Size

	Training	Evaluation	Testing	Total
Words	2,714,726	240,860	207,278	3,162,864
Named entities	246,529	23,176	18,315	288,020
Clauses	215,301	17,550	16,111	248,962
Sentences	63,310	5,620	5,250	74,180
Distinct words	42,091	(oov) 2,595	(oov) 2,006	46,692
Breaking spaces*	63,310	5,620	5,250	74,180
Non-breaking spaces**	402,380	39,920	32,204	475,504

* Breaking space = space that is used as a sentence boundary marker

** Non-breaking space = space that is not used as a sentence boundary marker

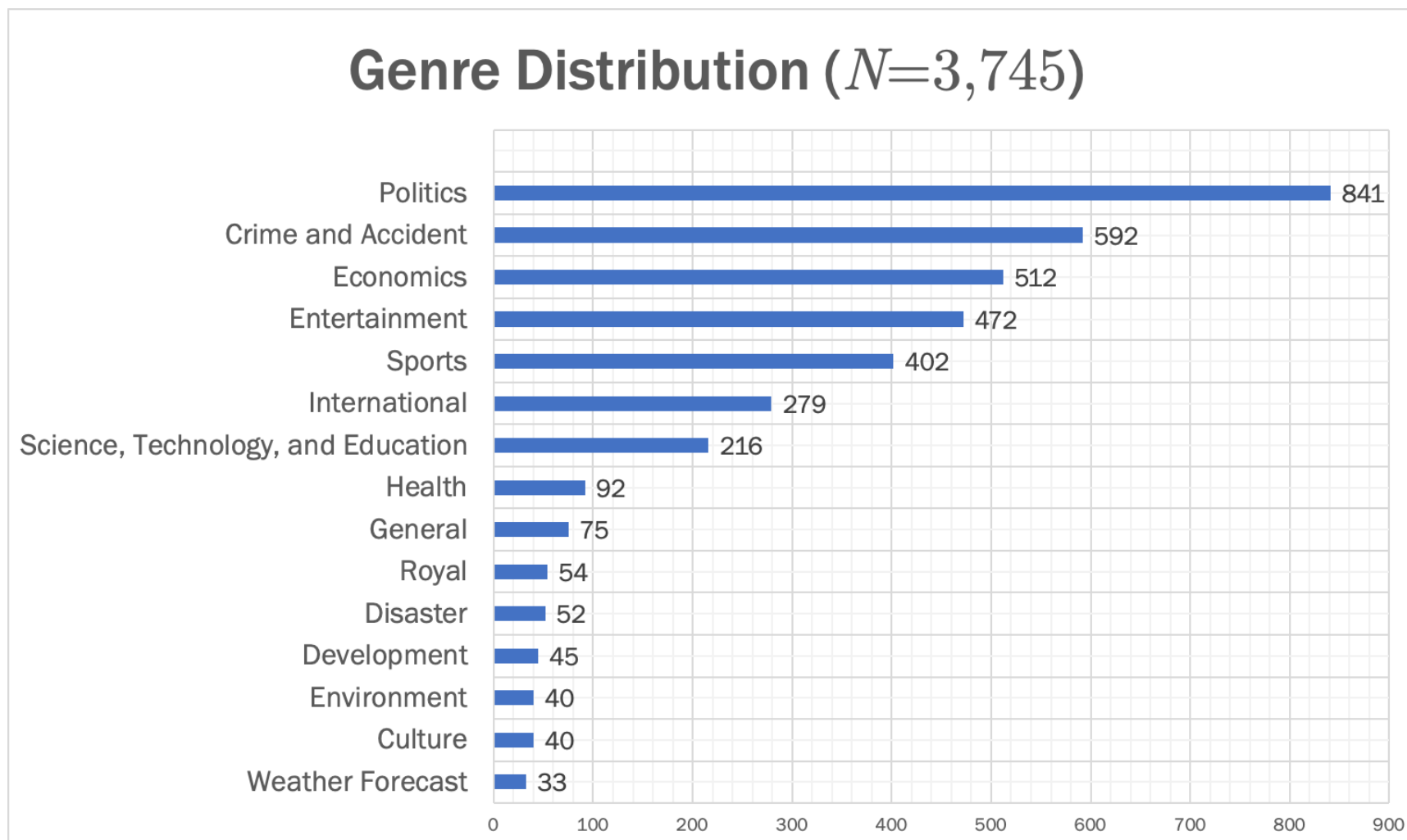
Experiment Results

- ระบบกำกับข้อมูลทางไวยากรณ์ของคำโดยอัตโนมัติ
 - แบ่งข้อมูล 3 ล้านคำออกเป็น train-dev-test ด้วยอัตราส่วน 8-1-1
 - ใช้เวกเตอร์ของคำและตัวอักษรขนาด 32 มิติ
 - ทรนโดยแบ่งข้อมูลชุดละ 16 ประโยคแบบสุ่มลำดับ และทรนเวียนซ้ำ 10 รอบใหญ่

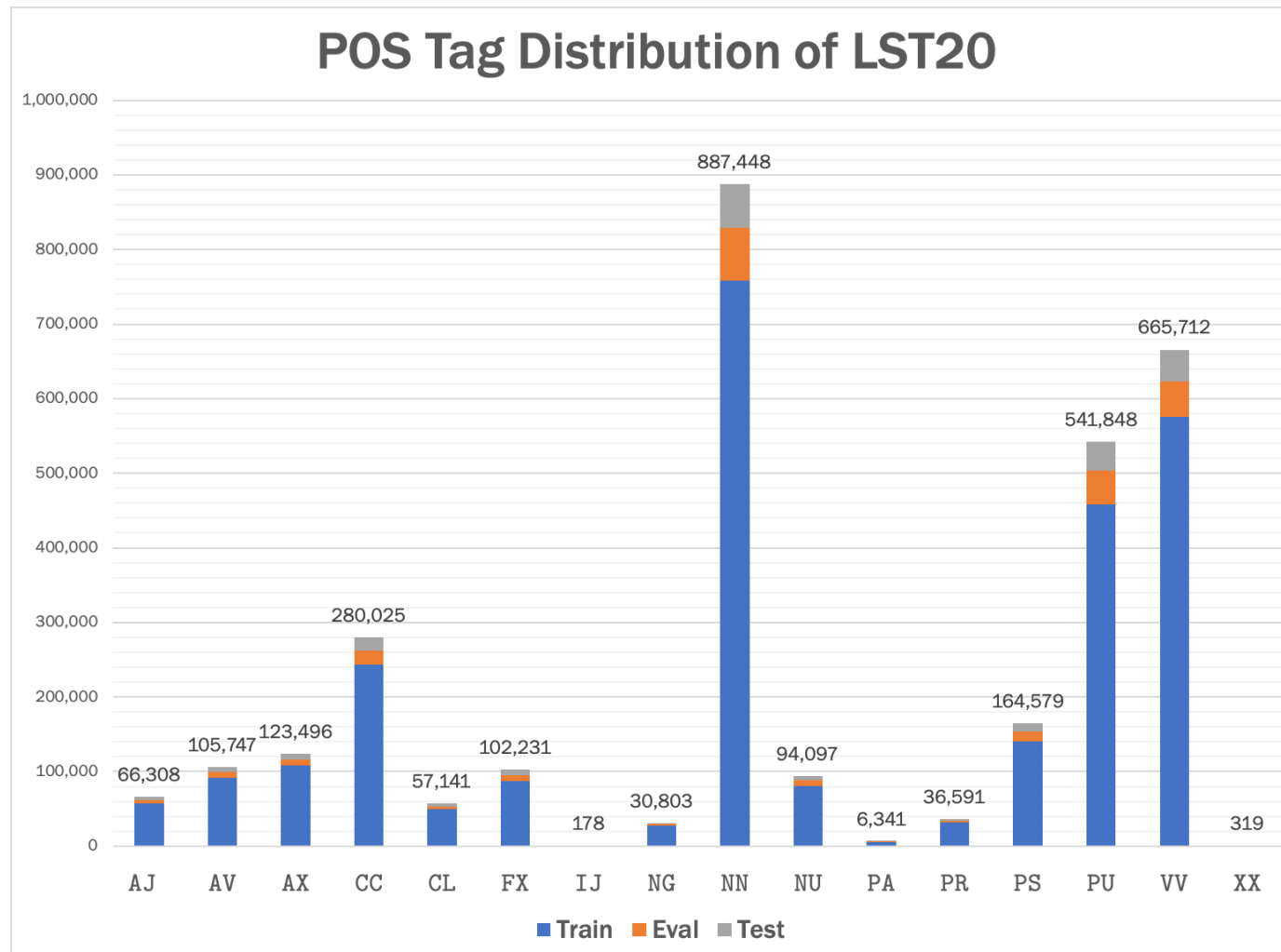
	%Precision	%Recall	%F1	ระบบเดิม (%F1)
การตัดคำ	96.51	98.13	<u>97.32</u>	94.18
การกำกับชนิดของคำ	92.77	94.33	<u>93.54</u>	92.78
การกำกับนิพจน์ระบุนาม	70.76	68.05	69.37	<u>71.30</u>
การตัดประโยคย่อย	82.42	67.13	<u>67.49</u>	ไม่เคยมีมาก่อน
การตัดประโยคใหญ่	79.35	100.00	<u>88.49</u>	ไม่เคยมีมาก่อน
การทายคำไม่รู้จัก	—	68.05	—	—

เตรียมนำระบบนี้เข้าสู่
แพลตฟอร์ม AI for
Thai ในต้นปี 2565

Statistics: Genres of News Articles



Statistics: POS Distribution



Thank You